Orchestrator runtime flow

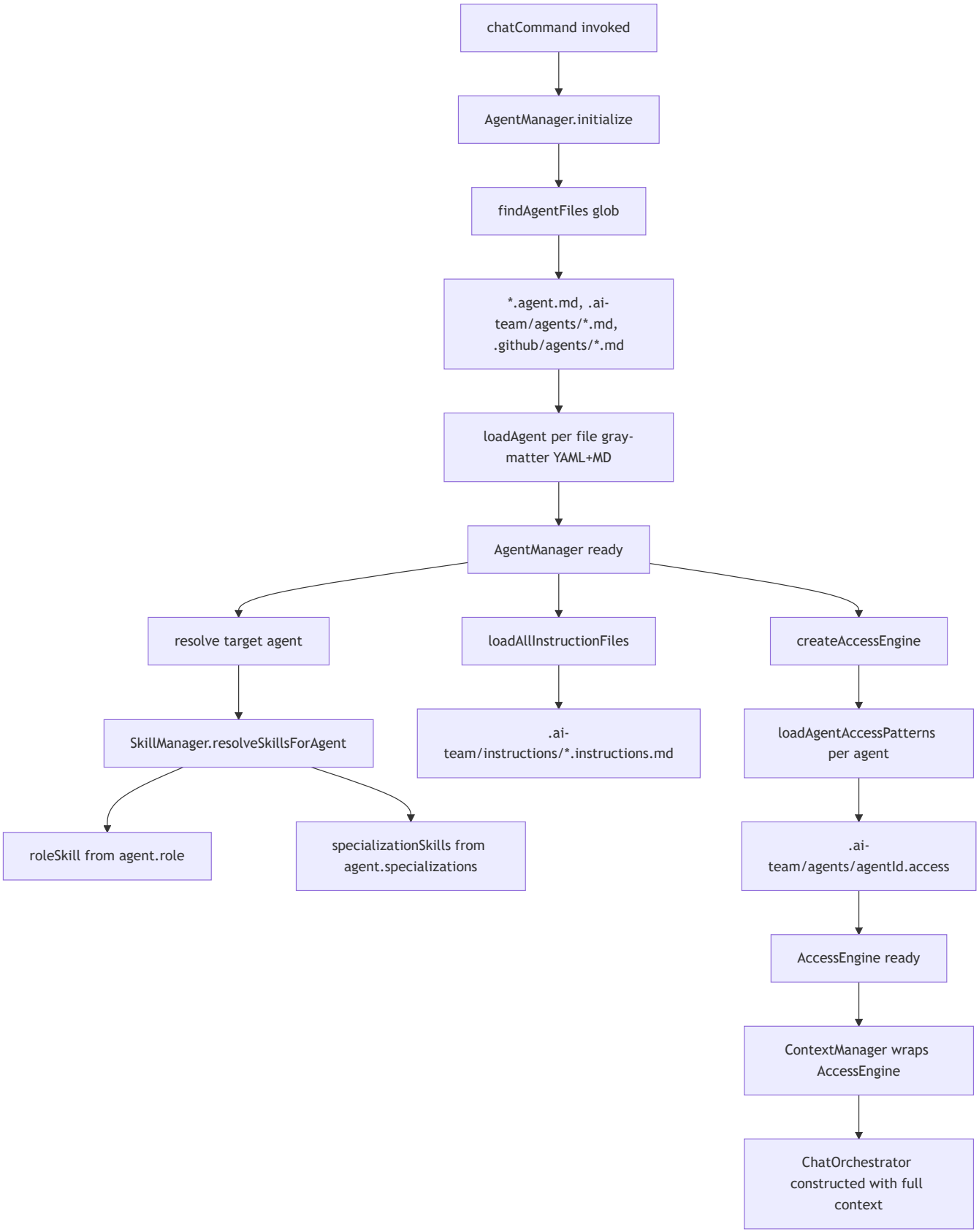
Scope: technical flow inside `@ai-team/service` after a message is submitted.

Components and responsibilities

- `chat-orchestrator.ts`: entry loop, pre-turn interceptors, hop control, handoff/hire handling
- `send-turn.ts`: one LLM turn (persist, build context, resolve skills, call model, run hook plugins, parse outcomes)
- `tool-dispatch.ts`: single tool gate (confirm, policy, execute, emit tool events)
- `handoff.ts`: session resolution + briefing + context switch
- `stream-events.ts`: runtime event wrappers over `hooks.emit`
- `pipeline.ts`: all extension-surface interfaces (open/closed principle)
- `defaults/hook-plugins.ts`: default `IOrchestratorHookPlugin` implementations including `StripInternalHandoffDirectivePlugin`

Startup phase (before first message)

Before any message reaches the orchestrator, `chatCommand()` runs a wired startup sequence.



What each piece provides

Component	File pattern	Provides
Agent portfolio	`**/*.agent.md`, `.ai-team/agents/*.md`, `.github/agents/*.md`	Identity, role, reporting line, `specializations`

Component	File pattern	Provides
Role skill instructions	`.ai-team/roles/<role>.md`	System prompt body for the agent's role (resolved by `SkillManager` via `agent.role`)
Specialization skill instructions	`.ai-team/roles/<specialization>.md`	Additional system prompt sections for each value in `agent.specializations[]`
Workspace instructions	`.ai-team/instructions/*.instructions.md`	Appended instructions filtered by `applyTo` glob
Per-agent access rules	`.ai-team/agents/<agentId>.access`	File-level read/write/create/delete permission patterns
Global fileTree config	`config.json → fileTree`	Workspace-wide path overrides merged with agent rules

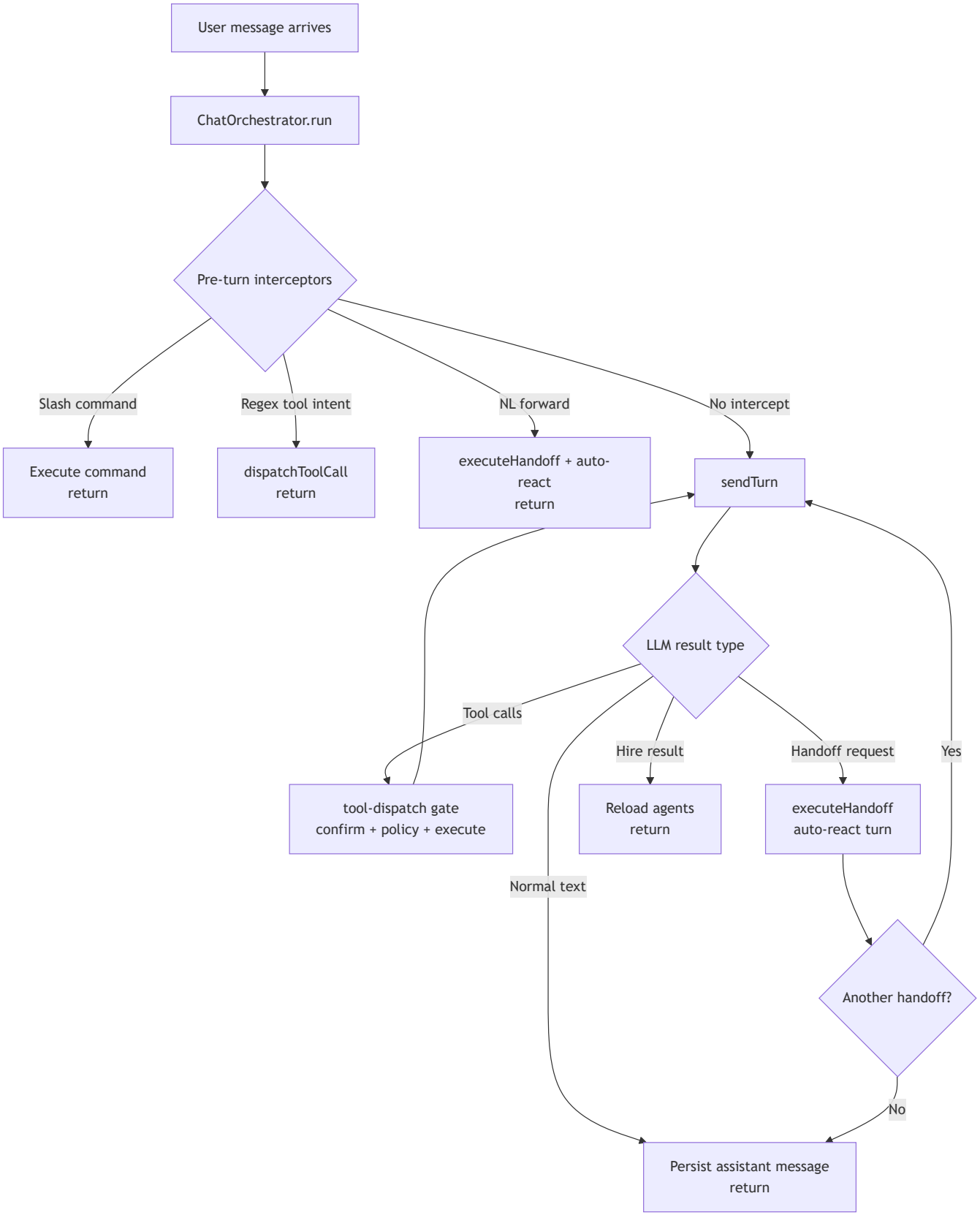
Access enforcement

`ContextManager.canRead/canWrite/canCreate/canDelete` delegates to `AccessEngine`. Every file tool call goes through `ContextManager` before execution. The engine evaluates the requesting agent against the pattern set from its `.access` file.

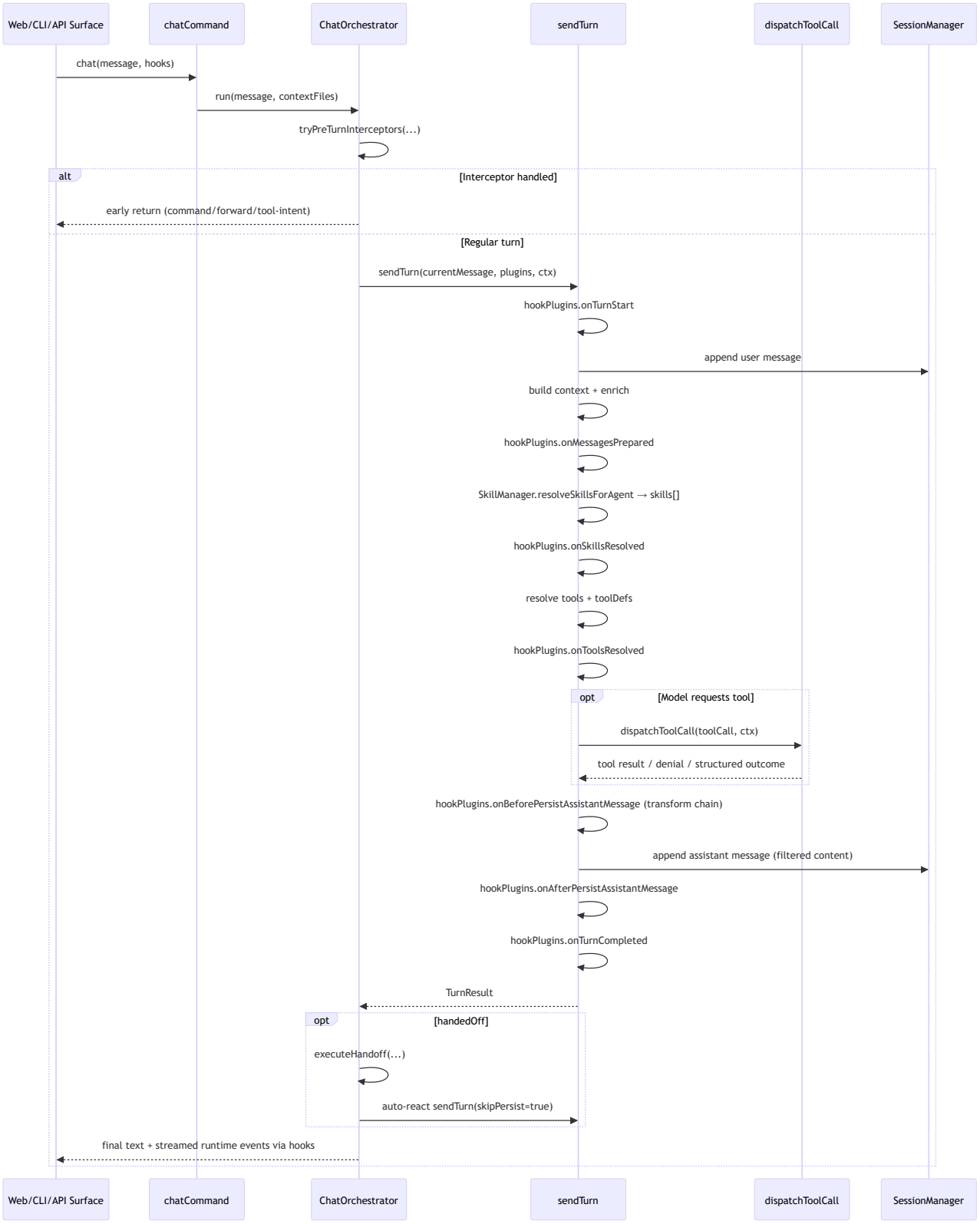
```
// packages/core/src/context/index.ts
canRead(agent: Agent, filePath: string): AccessVerdict {
  return this.engine
    ? this.engine.evaluate(agent.id, 'read', filePath)
    : this.legacyGlobCheck(agent, filePath);
}
```

```
// packages/core/src/storage/index.ts — loading access patterns at startup
export async function loadAgentAccessPatterns(workspaceRoot: string, agent
Id: string): Promise<AccessPatternSet> {
  const filePath = getAgentAccessFilePath(workspaceRoot, agentId); // .ai-
team/agents/<id>.access
  const content = await fs.readFile(filePath, 'utf-8');
  const rules = parseAccessFile(content);
  return accessRulesToPatternSet(rules);
}
```

Control flow (single message)



Information flow (sequence)

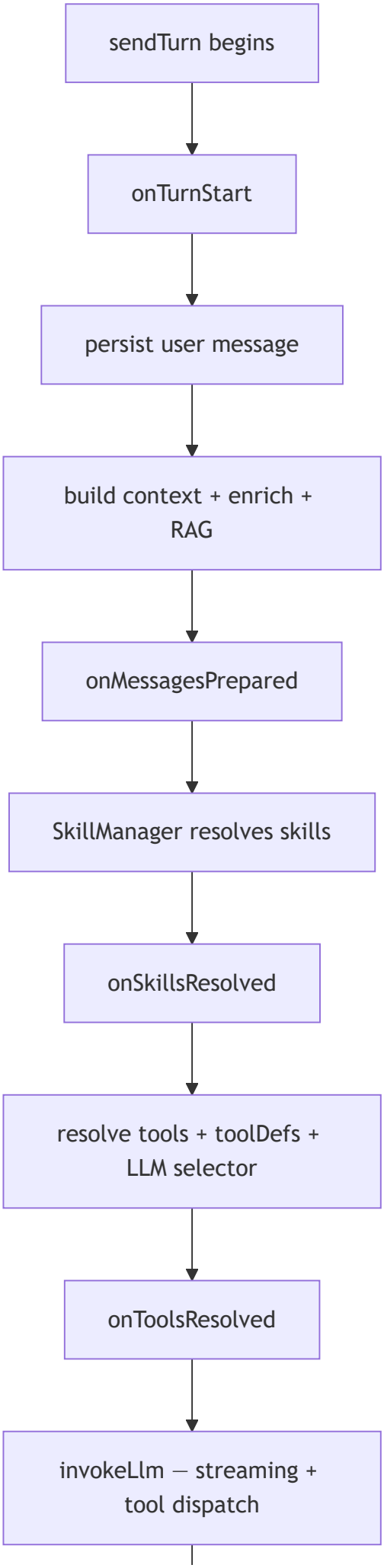


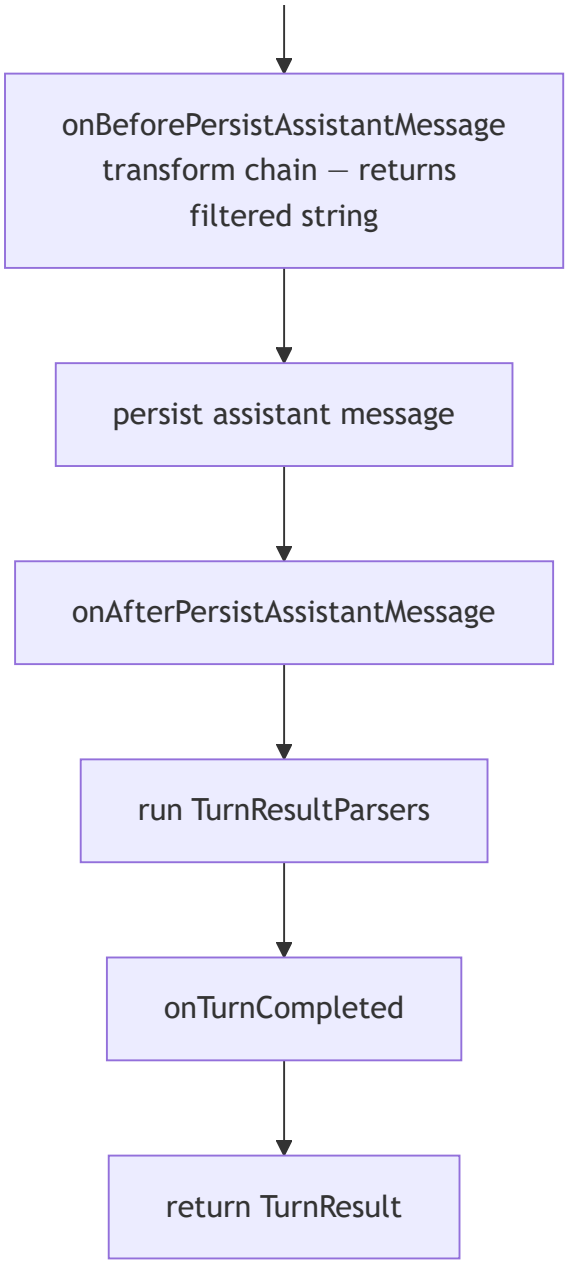
Hook plugin lifecycle

The orchestrator exposes a multi-hook plugin extension point via `IOrchestratorHookPlugin` (defined in `pipeline.ts`). Each plugin implements any subset of the 7 lifecycle hooks; all hooks are optional.

Hook plugins are registered in the DI container under `TOKENS.HookPlugins` and resolved into `ResolvedPlugins.hookPlugins`. The built-in set is created by `buildDefaultHookPlugins()` in

defaults/hook-plugins.ts.


sendTurn begins
onTurnStart
persist user message
build context + enrich + RAG
onMessagesPrepared
SkillManager resolves skills
onSkillsResolved
resolve tools + toolDefs + LLM selector
onToolsResolved
invokeLlm — streaming + tool dispatch



Hook contracts

Hook	Type	Payload	Return
`onTurnStart`	void	`{ userMessage, options?, ctx }`	—
`onMessagesPrepared`	void	`{ messages, ctx }`	—
`onSkillsResolved`	void	`{ skills[], missingSkillNames[], ctx }`	—
`onToolsResolved`	void	`{ tools[], toolDefs[], ctx }`	—
`onBeforePersistAssistantMessage`	**transform**	`{ fullResponse, persistedContent, ctx }`	`string | void` — returned string replaces

Hook	Type	Payload	Return
			`persistedContent` for persistence
`onAfterPersistAssistantMessage`	void	`{ fullResponse, persistedContent, persistedMessage, ctx }`	—
`onTurnCompleted`	void	`{ fullResponse, persistedContent, structuredResults, turnResult, ctx }`	—

`onBeforePersistAssistantMessage` is a **transform chain**: each plugin in sequence receives the `persistedContent` from the previous plugin and returns an updated string (or `void`/`undefined` to pass through unchanged). The final value is what gets written to the database — the developer never sees the raw `fullResponse` bearing internal directives.

Default hook plugins

Plugin class	Hook	Purpose
`StripInternalHandoffDirectivePlugin`	`onBeforePersistAssistantMessage`	Strips `HANDOFF:` / `FORWARD_TO:` directives from the persisted content. Emits an `info` log event when stripping occurs.

Registering a custom plugin

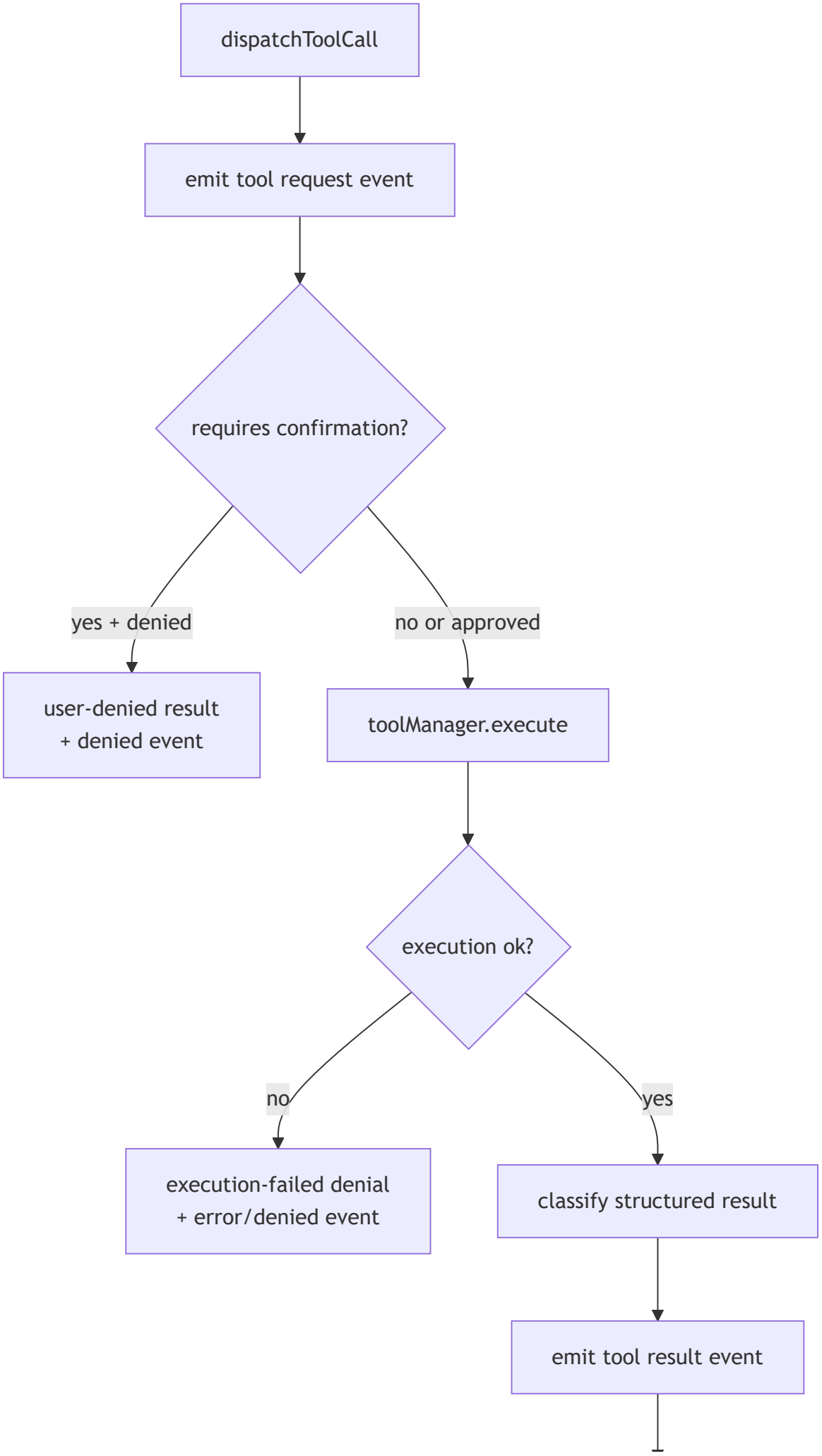
Implement `IOrchestratorHookPlugin` and pass it in `OrchestratorPlugins.hookPlugins[]`. Custom plugins are merged with the default set; they do not replace it.

```
class MyAuditPlugin implements IOrchestratorHookPlugin {
  readonly name = 'my-audit';

  onTurnCompleted({ turnResult, ctx }: TurnCompletedHookPayload): void {
    // log to external audit sink
  }
}

chatCommand({ hookPlugins: [new MyAuditPlugin()] });
```

Tool gate details


dispatchToolCall
emit tool request event
requires confirmation?
yes + denied
no or approved
user-denied result
+ denied event
toolManager.execute
execution ok?
no
yes
execution-failed denial
+ error/denied event
classify structured result
emit tool result event

Simplified implementation snippets (close to real code)

1) Interceptor short-circuit

```
const preTurnResult = await this.tryPreTurnInterceptors(message, options.c
ontextFiles);
if (preTurnResult !== undefined) return preTurnResult;
```

2) Pre-turn interceptor structure (simplified, close to real)

```
private async tryPreTurnInterceptors(message: string, contextFiles?: strin
g[]) {
  const slashResult = await this.trySlashCommand(message);
  if (slashResult !== null) return slashResult;

  const regexIntentResult = await this.tryRegexToolIntent(message, context
Files);
  if (regexIntentResult !== null) return regexIntentResult;

  const nlResult = await tryNlForward(message, this.ctx);
  if (nlResult === null) return undefined;

  if (nlResult === 'forwarded') {
    await sendTurn(HANDOFF_AUTO_MSG, this.plugins, this.ctx, { skipPersis
t: true });
    return '';
  }

  return nlResult;
}
```

3) Main turn loop and outcome handling

```
for (let hops = 0; hops < maxHops; hops++) {
  const result = await sendTurn(currentMessage, this.plugins, this.ctx);
  lastText = result.text;
```

```
  if (result.hired) {
    await this.ctx.agentManager.loadAllAgents();
    break;
  }

  if (result.handedOff && result.handoffTargetId) {
    const switched = await executeHandoff(
      this.ctx,
      result.handoffTargetId,
      result.handoffTargetSessionId,
      result.handoffNote,
    );
    if (!switched) break;

    const autoResult = await sendTurn(autoMsg, this.plugins, this.ctx, { s
kipPersist: true });
    lastText = autoResult.text;
    if (!autoResult.handedOff) break;
    continue;
  }

  break;
}
```

4) sendTurn skill resolution (via SkillManager)

```
const resolvedSkills = ctx.skillManager.resolveSkillsForAgent(ctx.agent);
if (resolvedSkills.roleSkill) {
  emitLog(hooks, 'info', `[skills] Loaded role skill: ${resolvedSkills.rol
eSkill.name}`);
}
for (const skill of resolvedSkills.specializationSkills) {
  emitLog(hooks, 'info', `[skills] Loaded specialization skill: ${skill.na
me}`);
}
for (const missing of resolvedSkills.missingSkillNames) {
  emitLog(hooks, 'warn', `[skills] Skill not found: ${missing}`);
}
// resolvedSkills.skills is the merged Skill[] passed to llmService.chatWi
thTools(...)
```

`resolveSkillsForAgent` returns `ResolvedAgentSkills` (exported from `@ai-team/core/skill`):

```
interface ResolvedAgentSkills {
  roleSkill?: Skill;               // matched by agent.role
  specializationSkills: Skill[];   // matched by agent.specializations[]
  skills: Skill[];                 // merged: [roleSkill?, ...specializati
onSkills]
  missingSkillNames: string[];     // skill names that were not found in t
he skill library
}
```

5) sendTurn tool-calling callback into secure dispatch

```
llmService.chatWithTools(..., async (toolCall) => {
  const response = await dispatchToolCall(
    { toolCallId: toolCall.toolCallId, toolName: toolCall.toolName, args:
toolCall.args },
    ctx,
  );
  return {
    toolCallId: response.toolCallId,
    toolName: response.toolName,
    result: response.result,
    isError: response.isError,
  };
});
```

6) sendTurn hook plugin transform chain (before persistence)

```
const persistedContent = await runBeforePersistMessageHooks(
  hookPlugins,
  { fullResponse, persistedContent: fullResponse, ctx },
  hooks,
);
// Each plugin in hookPlugins that implements onBeforePersistAssistantMess
age
// receives the current persistedContent and may return a modified string.
// The final value is written to the session store.
```

7) tool-dispatch execution and eventing (simplified)

```
emitEvent(ctx.hooks, { kind: 'tool', toolName, toolPhase: 'request', messa
ge: label });

const deniedByUser = await requestExecutionApproval(toolName, label, ctx);
if (deniedByUser) {
  return { toolCallId, toolName, result: deniedByUser.message, isError: fa
lse, denial: deniedByUser };
}

const execResult = await ctx.toolManager.execute(
  ctx.agent,
  toolName,
  args,
  { workspaceRoot: ctx.workspaceRoot, currentFiles: contextFiles },
);

emitToolEvent(ctx.hooks, toolName, execResult.ok ? 'result' : 'error',
...);
```

8) Runtime event helper usage

```
emitStatus(hooks, 'thinking');
emitStatus(hooks, 'handoff', `${this.ctx.agent.name} taking over.`);
emitLog(hooks, 'warn', 'Handoff requested to unknown agent');
```

Runtime events emitted

The orchestrator path emits structured runtime events through `hooks.emit`:

- `status`: phase changes (`thinking`, `handoff`, `error`)
- `token`: streamed text deltas
- `tool`: request/start/result/error/denied
- `question`: confirmation/input/checklist/select prompts
- `handoff`: from/to agent + session metadata
- `log`: runtime diagnostics (including `[skills]` events for loaded/missing skills and `[filter]` events from hook plugins)

Pipeline extension interfaces

All extension seams are defined in `pipeline.ts` (open/closed principle). The interfaces available as of this writing:

#	Interface	Role in the pipeline
1	`IContextCompressor`	Prune/summarize message history before context assembly
2	`IContextBuilder`	Assemble `ChatCompletionMessageParam[]` including system prompt
3	`IContextEnricher`	Role-aware system message injections (workspace overview, team roster, etc.)
4	`IRagProvider`	Retrieval-augmented generation — inject relevant file content
5	`IToolResolver`	Determine which tools are available to the agent for this turn
6	`IMcpGateway`	Discover tools from external Model Context Protocol servers
7	`ILlmSelector`	Select and initialize the LLM model/provider for the turn
8	`IOutputHandler`	Persist result and emit surface events
9	`ITurnResultParser`	Interpret raw turn outputs (handoff, hire, normal) — first match wins
10	`ISlashCommand`	A single `/key` command executed by the pre-turn interceptor
11	`IOrchestratorHookPlugin`	Multi-hook plugin: one plugin registers any subset of the 7 lifecycle hooks

Default implementations live under `orchestrator/defaults/`. Stub no-op implementations satisfy every interface so the pipeline is always fully wired.

Source files

- `packages/service/src/orchestrator/chat-orchestrator.ts`
- `packages/service/src/orchestrator/send-turn.ts`
- `packages/service/src/orchestrator/tool-dispatch.ts`
- `packages/service/src/orchestrator/handoff.ts`
- `packages/service/src/orchestrator/pipeline.ts` — all extension interfaces and payload types
- `packages/service/src/orchestrator/defaults/hook-plugins.ts` — default hook plugin implementations
- `packages/service/src/contracts.ts`
- `packages/core/src/skill/index.ts` — `SkillManager` + `resolveSkillsForAgent()` + `ResolvedAgentSkills`

Appendix: Source Files

packages/service/src/orchestrator/chat-orchestrator.ts

```
/**
 * ChatOrchestrator — the stateful session controller.
 *
 * One instance per active session. Accepts a single user message, runs
 * through the pipeline, handles handoff / hire / slash commands, and
 * returns when the turn is complete.
 *
 * The orchestrator itself knows nothing about:
 *   - HTTP, WebSockets, CLI readline — those are in the adapter layer.
 *   - Specific tool implementations — those are in core / service tools.
 *   - Storage format — delegated to SessionManager + ChatRuntimeHooks.
 *
 * To extend: register pipeline plugins via OrchestratorPlugins before cal
ling run().
 */

import { emitStatus, emitLog } from './stream-events.js';
import { resolvePreLlmIntent } from '../tools/pre-llm-intents.js';
import { sendTurn } from './send-turn.js';
import { executeHandoff, tryNlForward } from './handoff.js';
import { dispatchToolCall } from './tool-dispatch.js';
import type { OrchestratorContext } from './pipeline-context.js';
import type { ResolvedPlugins } from './pipeline.js';

/** Options for a single run() call. */
export interface RunOptions {
  /** Raw user message (may start with a slash command). */
  message: string;
  /** Files the user has open / selected in the IDE. */
  contextFiles?: string[];
  /**
   * Maximum number of automatic handoff hops allowed before the loop stop
s
   * and returns control to the caller. Prevents infinite handoff chains.
   * Default: 10.
   */
  maxHops?: number;
}

export class ChatOrchestrator {
```

```
  constructor(
    private readonly ctx: OrchestratorContext,
    private readonly plugins: ResolvedPlugins,
  ) {}

  /**
   * Execute one user turn. Returns the final response text.
   *
   * The loop continues across handoffs (agent switches) until:
   *   - The responding agent does NOT request a handoff, OR
   *   - The maximum hop count is reached.
   */
  async run(options: RunOptions): Promise<string> {
    const { message, maxHops = 10 } = options;

    const preTurnResult = await this.tryPreTurnInterceptors(message, optio
ns.contextFiles);
    if (preTurnResult !== undefined) return preTurnResult;

    // ── Turn loop (handles handoff chains) ──────────────────────────────
─────
    let currentMessage = message;
    let lastText = '';

    for (let hops = 0; hops < maxHops; hops++) {
      const result = await sendTurn(currentMessage, this.plugins, this.ct
x);
      lastText = result.text;

      // ── Hire: reload agents, notify surface, continue ─────────────────
─────
      if (result.hired) {
        await this.ctx.agentManager.loadAllAgents();
        emitLog(
          this.ctx.hooks,
          'info',
          `${this.ctx.agent.name} hired ${result.hired.name} (${result.hir
ed.role}).`,
        );
        // Hiring is a side-effect — the turn is still considered complet
e.
        break;
      }

      // ── Handoff: switch agent, auto-react ─────────────────────────────
──────
```

```
      if (result.handedOff && result.handoffTargetId) {
        const switched = await executeHandoff(
          this.ctx,
          result.handoffTargetId,
          result.handoffTargetSessionId,
          result.handoffNote,
        );

        if (!switched) {
          emitLog(
            this.ctx.hooks,
            'warn',
            `Handoff requested to unknown agent "${result.handoffTargetI
d}" — staying with ${this.ctx.agent.name}.`,
          );
          break;
        }

        emitStatus(
          this.ctx.hooks,
          'handoff',
          `${this.ctx.agent.name} taking over.`,
        );

        // The new agent has the LLM briefing in their session history.
        // Auto-run a turn so the recipient reacts to the handoff context
        // and asks the developer how to proceed.  skipPersist keeps the
        // synthetic prompt out of the DB.
        const autoMsg = `[Handoff received] You have just been handed this
conversation. `
          + `Review the briefing above, acknowledge the context, and ask t
he developer how they would like to proceed.`;
        const autoResult = await sendTurn(autoMsg, this.plugins, this.ctx,
{ skipPersist: true });
        lastText = autoResult.text;
        // If the auto-react itself triggers another handoff, the next
        // iteration of the loop will handle it.
        if (!autoResult.handedOff) break;
        continue;
      }

      // Normal turn — no handoff, we're done.
      break;
    }

    return lastText;
```

```
  }

  private async tryPreTurnInterceptors(
    message: string,
    contextFiles?: string[],
  ): Promise<string | undefined> {
    // ── Slash command intercept ──────────────────────────────────────
─────
    const slashResult = await this.trySlashCommand(message);
    if (slashResult !== null) return slashResult;

    // ── Deterministic regex tool intents (pre-LLM) ──────────────────
────
    const regexIntentResult = await this.tryRegexToolIntent(message, conte
xtFiles);
    if (regexIntentResult !== null) return regexIntentResult;

    // ── Natural-language forward detection ──────────────────────────
─────
    const nlResult = await tryNlForward(message, this.ctx);
    if (nlResult === null) return undefined;

    if (nlResult === 'forwarded') {
      // Handoff succeeded — auto-run a turn so the new agent reacts to
      // the briefing and asks the developer how to proceed.
      const autoMsg = `[Handoff received] You have just been handed this c
onversation. `
        + `Review the briefing above, acknowledge the context, and ask the
developer how they would like to proceed.`;
      await sendTurn(autoMsg, this.plugins, this.ctx, { skipPersist: true
});
      return '';
    }

    return nlResult;
  }

  // ── Slash command handling ──────────────────────────────────────────
──────

  /**
   * Returns the response string if a slash command was matched, null othe
rwise.
   */
  private async trySlashCommand(message: string): Promise<string | null> {
    const trimmed = message.trim();
```

```
    if (!trimmed.startsWith('/')) return null;

    const [rawKey, ...rest] = trimmed.slice(1).split(/\s+/);
    const key = (rawKey ?? '').toLowerCase();
    if (!key) {
      emitLog(this.ctx.hooks, 'warn', 'Please enter a slash command name.
Try /help.');
      return '';
    }
    const rawArgs = rest.join(' ');

    const command = this.plugins.slashCommands.find(
      c => c.key === key || c.aliases?.includes(key),
    );

    if (!command) {
      emitLog(this.ctx.hooks, 'warn', `Unknown command: /${key}. Try /hel
p.`);
      return '';
    }

    await command.execute(rawArgs, this.ctx);
    return '';   // Slash commands handle their own output via hooks / std
out.
  }

  /**
   * Deterministic tool intent layer checked before LLM turn execution.
   *
   * IMPORTANT: tool execution is routed through dispatchToolCall(), which
   * preserves policy enforcement and dangerous-tool confirmation prompts.
   */
  private async tryRegexToolIntent(message: string, contextFiles?: string
[]): Promise<string | null> {
    const intent = resolvePreLlmIntent(message);
    if (!intent) return null;

    await this.executeRegexToolIntent(intent.toolName, intent.args, contex
tFiles);
    return '';
  }

  private async executeRegexToolIntent(
    toolName: string,
    args: unknown,
    contextFiles?: string[],
```

```
  ): Promise<void> {
    await dispatchToolCall(
      {
        toolCallId: `regex-intent-${Date.now()}`,
        toolName,
        args,
      },
      this.ctx,
      contextFiles,
    );
  }

}
```